

Chapter 1: Concept and definition

1.1 Thermodynamics, Application of thermodynamics, Thermodynamic system, Macroscopic and Microscopic approaches

1.2 Properties and state of a substance: Thermodynamic properties and types, state, path, process, cycle.

1.3 Processes (definition, characteristics, features, Examples): Quasi-equilibrium process, reversible and irreversible process

1.4 Specific volume, Pressure, Atmospheric pressure, Gauge pressure, Absolute pressure

1.5 Equality of temperature: Zeroth law of thermodynamics

1.1 Thermodynamics, Application of thermodynamics, Thermodynamic system, Macroscopic and Microscopic approaches

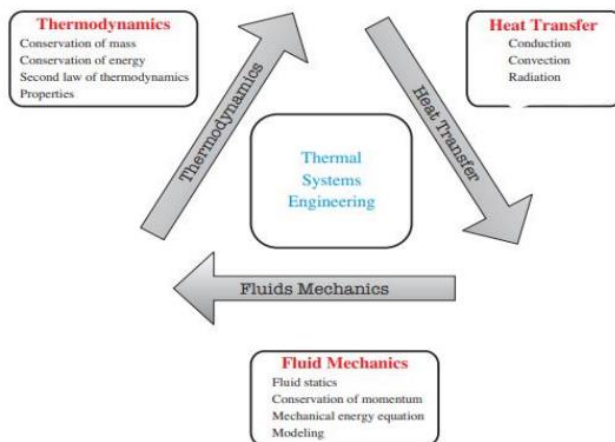
Thermodynamics

Thermal science is the combined study of thermodynamics, fluid mechanics, heat transfer

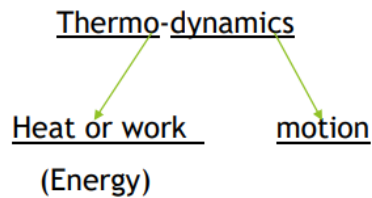
Thermodynamics: the study of energy, energy transformations and its relation to matter

Heat Transfer: the study of energy in transit including the relationship between energy, matter, space and time

Fluid Mechanics: the study of fluids at rest or in motion



Thermodynamics derived from two words 'Thermo' meaning 'Heat' or 'Work' and 'Dynamics' meaning 'Motion'.



Thus it is a science of energy transfer in form of heat and its effect on the physical properties of the substances

Application of thermodynamics

Thermodynamics has very wide applications as basis of thermal engineering. Almost all process and engineering industries, agriculture, transport, commercial and domestic activities use thermal engineering. But energy technology and power sector are fully dependent on the laws of thermodynamics.

For example:

- (i) Central thermal power plants, captive power plants based on coal.
- (ii) Nuclear power plants.
- (iii) Gas turbine power plants.
- (iv) Engines for automobiles, ships, airways, spacecrafts.
- (v) Direct energy conversion devices: Fuel cells, thermionic, thermoelectric engines.
- (vi) Air conditioning, heating, cooling, ventilation plants.
- (vii) Domestic, commercial and industrial lighting.
- (viii) Agricultural, transport and industrial machines.

All the above engines and power consuming plants are designed using laws of thermodynamics.

Computer engineer may find its application while selecting the proper cooling system for the different electrical and electronic devices

Heat engine and Refrigerator are most common thermodynamic applications

- Heat engine: heat is continuously converted into work

- Refrigerator: heat is continuously transferred from low temperature to high temperature with the help of work input to produce the desired cooling effect

Thermodynamic system

The thermodynamics system may be broadly defined as a definite area or space where some thermodynamics process is taking place. It may be noted that a thermodynamics system has its boundaries and anything outside the boundaries is called surroundings

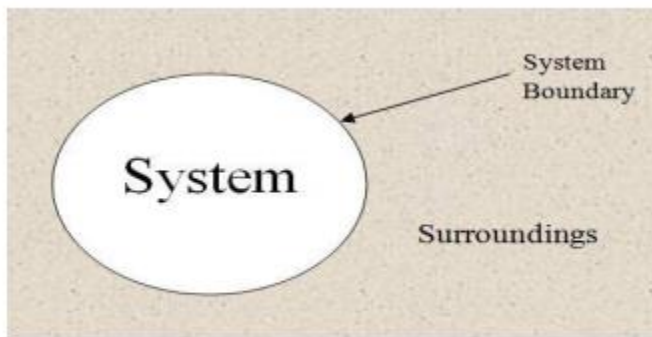
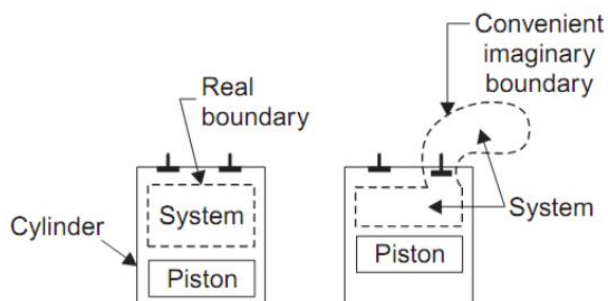


Fig- Thermodynamic system

The actual or hypothetical envelope enclosing the system is the boundary of the system.



The real and imaginary boundaries.

Universe = system + surrounding

Types of thermodynamic system

- a) **Isolated system:** These systems cannot have either energy or mass transfer with the surroundings.
- b) **Control mass or closed system:** Across the boundary of a closed system the transfer of energy (Work and/or heat) takes place but transfer of mass does not take place. E.g.: Piston and cylinder.

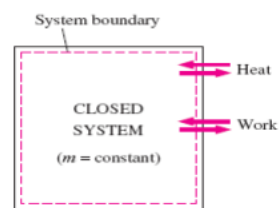
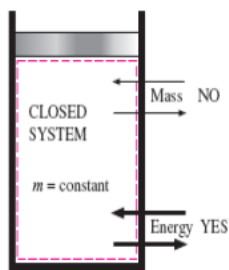


FIGURE 2-1 1
Energy can cross the boundaries of a closed system in the form of heat and work.

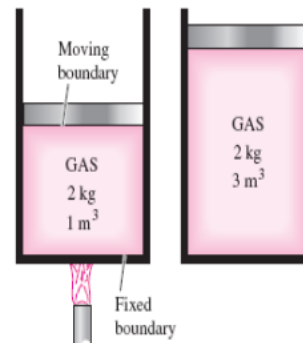
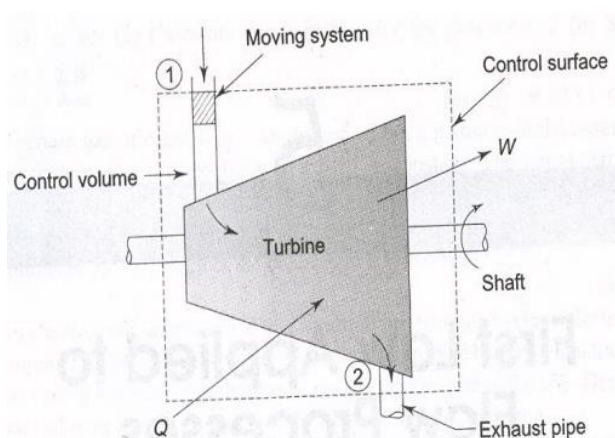
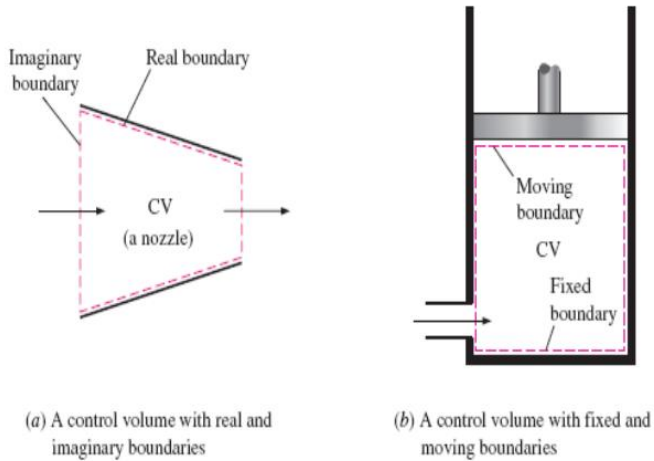


Fig. Closed System

Fig. A Closed System with moving boundary

- c) **Control volume or open system:** In this system mass and energy both may be transferred between the system and the surroundings. E.g.: Gas turbine, Steam turbine, Compressor, Boiler etc.





- d) **Adiabatic System** : An adiabatic system is one which is thermally insulated from its surroundings. It can, however, exchange work with its surroundings. If it does not, it becomes an isolated system.
- e) **Homogeneous System** A system which consists of a single phase is termed as homogeneous system. Examples: Mixture of air and water vapour, water plus nitric acid and octane plus heptane.
- f) **Heterogeneous System** :A system which consists of two or more phases is called a heterogeneous system. Examples: Water plus steam, ice plus water and water plus oil.

Macroscopic and Microscopic approaches

Thermodynamic studies are undertaken by the following two different approaches.

1. Macroscopic approach (Macro means big or total)
2. Microscopic approach (Micro means small)

These approaches are discussed (in a comparative way) below:

S. No.	Macroscopic approach	Microscopic approach
1.	In this approach a certain quantity of matter is considered <i>without</i> taking into account the events occurring at molecular level. In other words this approach to thermodynamics is concerned with <i>gross or overall behaviour</i> . This is known as <i>classical thermodynamics</i> .	The approach considers that the system is made up of a very large number of discrete particles known as <i>molecules</i> . These molecules have different velocities and energies. The values of these energies are constantly changing with time. This approach to thermodynamics which is concerned directly with the <i>structure of the matter</i> is known as <i>statistical thermodynamics</i> .
2.	The analysis of macroscopic system requires simple mathematical formulae.	The behaviour of the system is found by using statistical methods as the number of molecules is very large. So advanced statistical and mathematical methods are needed to explain the changes in the system.
3.	The values of the properties of the system are their average values. For example, consider a sample of a gas in a closed container. The <i>pressure</i> of the gas is the average value of the pressure exerted by millions of individual molecules. Similarly the <i>temperature</i> of this gas is the average value of translational kinetic energies of millions of individual molecules. These properties like <i>pressure and temperature</i> can be measured very easily. <i>The changes in properties can be felt by our senses</i> .	The properties like <i>velocity, momentum, impulse, kinetic energy, force of impact</i> etc. which describe the molecule <i>cannot be easily measured by instruments. Our senses cannot feel them</i> .
4.	In order to describe a system only a few properties are needed.	Large number of variables are needed to describe a system. So the approach is complicated.

1.2 Properties and state of a substance: Thermodynamic properties and types, state, path, process, cycle.

Thermodynamic properties

The variables which determine state or exact condition of a substance or system is called as its properties. The various properties of thermodynamic system are pressure, temperature, specific volume, internal energy, enthalpy, entropy, etc.

OR

The state of a system may be identified by certain observable quantities such as volume, temperature, pressure and density etc. All the quantities which identify the state of a thermodynamic system are called thermodynamic properties.

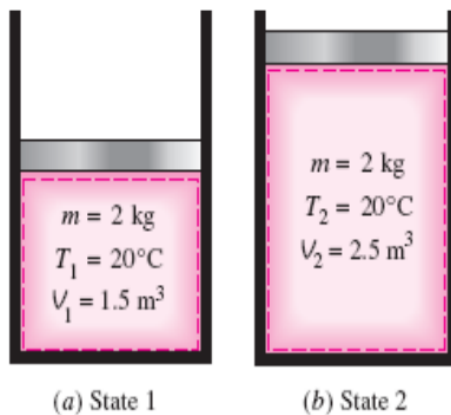
Thermodynamic state

Each unique condition at which a thermodynamic system can exist is called a thermodynamic state.

Example: At any instant if room has temp. of 27deg and pressure of 1atm this condition of a system defined by $T_1 = 27^\circ\text{C}$ and $P_1 = 1\text{atm}$ can be specified as state 1 of the system.

OR

Exact condition of a system/substance is called state.



Thermodynamic properties can be divided into two groups:

1. Extensive properties
2. Intensive properties

1. Extensive properties:

Thermodynamic property which is dependent or proportional to mass or part of the system considered is called an extensive property.

OR

The properties of the system, whose value for the entire system is equal to the sum of their values for the individual parts of the system for example: Total volume, total mass and total energy

2. Intensive properties

Thermodynamic property which is independent of mass or part of the system considered is called an intensive property

OR

Whose value for the entire system is not equal to the sum of their values for the individual parts of the system. For eg. Temperature, pressure and density, Specific volume, heat and work etc.

Thermodynamic process

When one or more of the thermodynamic properties of a system change, we say that there is a change of state of the system. This change of state of a system is referred to as thermodynamic process.

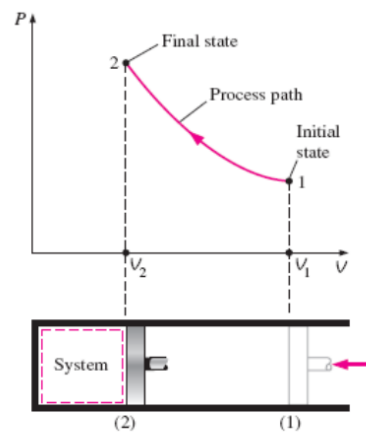
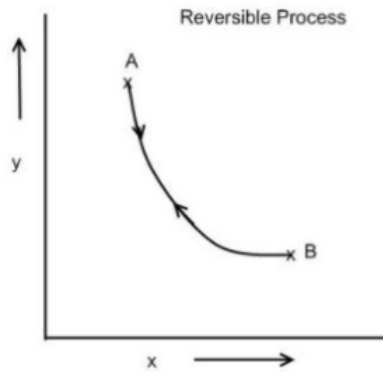


Fig. A Process between state 1 and state 2 and process path

Reversible and irreversible process:

A process is called a reversible process if it can be completely reversed and the system and the surroundings come back to its original states.

A process which cannot be exactly reversed and the system cannot be brought back to the initial state without leaving a net change in the surroundings is called irreversible process.

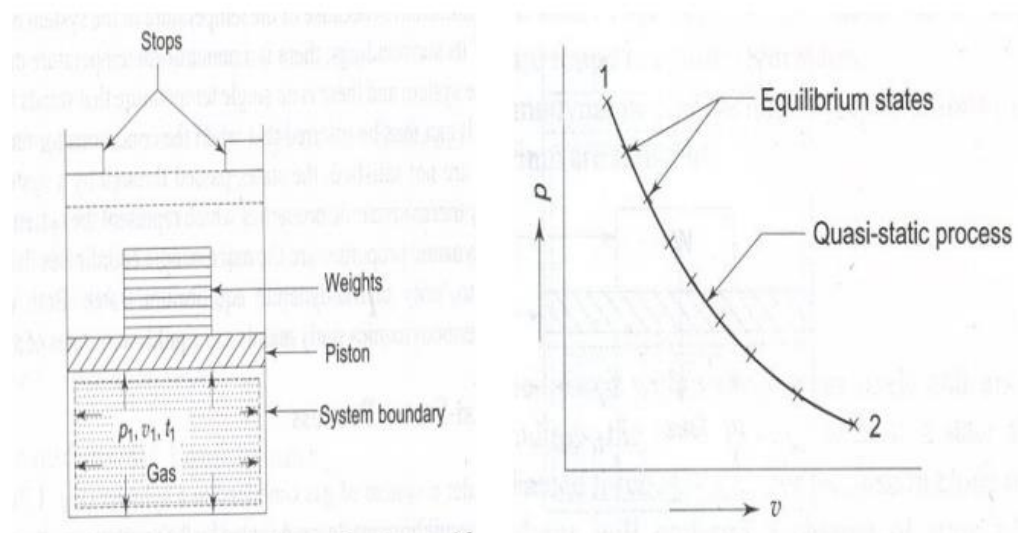


Quasi-static process (Quasi-equilibrium)

A quasi-static process is one in which the deviation from equilibrium is infinitesimally small. All the states through which a system passes during a process can be considered as a succession of equilibrium states.

OR

It is the process in which every state passed through by the system is an equilibrium state .



Consider a system of gas contained in a cylinder which initially is in equilibrium state represented by the properties (p_1, v_1, t_1) as shown in the figure

The weight on the piston just balance upward force exerted by gas, if total weight is removed, there will be unbalanced force between system and surrounding and the piston will move upward till it hits the stop.

System again comes to an equilibrium state represented by the properties (p_2, v_2, t_2)

But the intermediate state passed through by the system are non-equilibrium which cannot be described by the thermodynamic coordinates

Now if weight on piston is made up of many pieces of infinitesimally small weight and these weights are removed one by one very slowly from top of the piston, the piston will move upward very slowly.

Now the unbalanced force between system and surrounding would be infinitesimally small and the system would be almost close to equilibrium state and hence every state passed by the system will be an equilibrium state and the locus of a series of such equilibrium states is called a quasi-static process.

Thus quasi-equilibrium process takes place very slowly. If one property remains constant during a process, the prefix “iso” is used to describe such a process.

E.g. Isothermal process – Constant temperature process

Isobaric process – Constant pressure process

Isochoric process – Constant volume process.

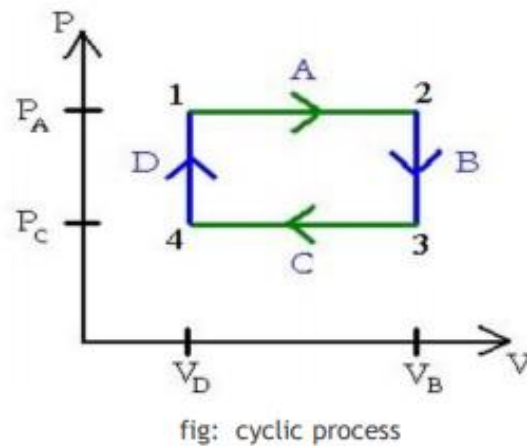
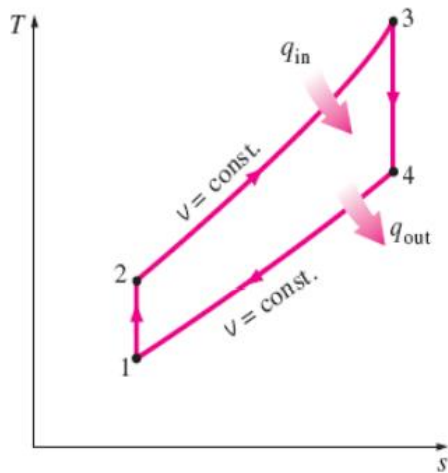
Thermodynamic cycle

When a number of processes in sequence bring the system back to its initial state, then the system is said to have undergone a thermodynamic cycle.

OR

The path followed by a system when it undergoes from one equilibrium state to another is called thermodynamic process.

Cyclic process: A process is said to be a cyclic process, if the initial state of the system is restored by a no. of different process in series.



In Fig 1-2-3-4-1 is cycle consisting of four processes. The change in value of any property is zero for a cyclic process.

Equilibrium

The system is said to be thermodynamic equilibrium if there is no change in any properties of the system when it is isolated from its surroundings.

Types of Thermodynamic Equilibrium:

- Chemical Equilibrium
- Mechanical Equilibrium and
- Thermal Equilibrium

Chemical Equilibrium

A system is said to be chemical equilibrium when no chemical reaction takes place in the system or between system and surrounding. The chemical composition will be the same throughout the system and this will not vary that system in chemical equilibrium. This is a reversible reaction.

Example: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ When we heat Calcium Carbonate (CaCO_3) at temperature 1073K will get CaO and CO_2

Mechanical Equilibrium

A system is said to be mechanical equilibrium when there is no unbalanced force within the system or between system and surroundings.

Thermal Equilibrium

A system is said to be thermal equilibrium when there is no temperature difference present and the temperature remains equal at all point.

Specific volume:

Specific volume of a substance is its volume per unit mass. Its unit is m³/kg. Density of a substance is its mass per unit volume. Unit - kg /m³.

$$\text{Density: } \rho = \frac{m}{V} \quad (\text{kg/m}^3)$$

Specific volume

$$v = \frac{V}{m} = \frac{1}{\rho}$$

Pressure

Pressure may be defined as normal force per unit area. In S.I system the unit of pressure is pascal (Pa).

$$1\text{Pa} = 1\text{N/m}^2; \quad 1\text{kPa} = 1000\text{Pa}; \quad 1\text{MPa} = 10^6\text{Pa} = 1000\text{kPa}$$

$$1 \text{ standard atmospheric pressure} = 1.01325\text{bar} = 101.325\text{kPa} = 101325\text{Pa} = 760\text{mm of Hg}$$

$$1\text{mm of Hg} = 0.1333\text{kPa} = 13.6\text{mm of H}_2\text{O}$$

$$1\text{bar} = 10^5\text{Pa} = 100\text{kPa}; \\ 1\text{N/mm}^2 = 10^6\text{N/m}^2 = 10^6\text{Pa} = 1000\text{kPa}$$

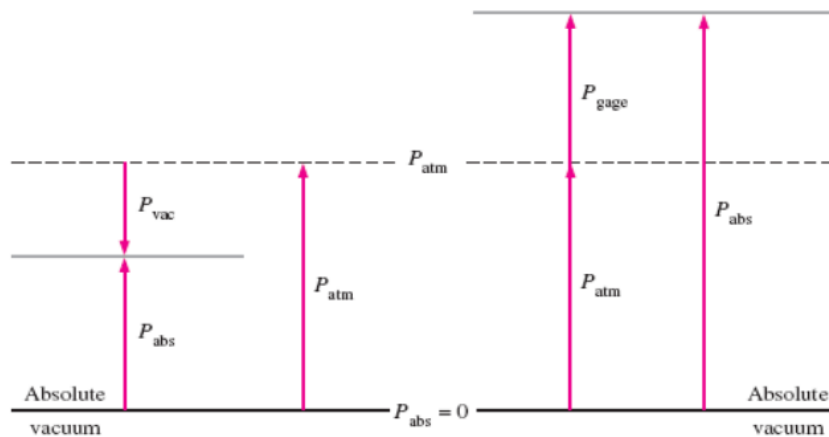
The pressure exerted by atmosphere on earth surface is known as atmospheric pressure, measured by barometer. At the sea level and at 0 °C, 760mm of Hg is known as standard atmospheric pressure. Gauges, Gauge pressure, Vacuum pressure, Absolute pressure

The actual pressure of the working substance is called the absolute pressure. For gauge pressure or positive pressure,

$$P_{\text{absolute}} = P_{\text{atm}} + P_{\text{gauge}}$$

For vacuum or negative pressure

$$P_{\text{absolute}} = P_{\text{atm}} - P_{\text{gauge}}$$



Equality of temperature

Two systems are said to have equal temperature, if there is no change in any properties of both the system when they are brought in contact with each other



fig: An arrangement for equality of temperature

Zeroth law of thermodynamic

